Ex. No. : 3 IMPLEMENTATION OF DEADLOCKS Date: 16.12.2025

AIM

To implement Banker's Algorithm for deadlock avoidance.

PROCEDURE

1. Define allocation, maximum, and available resource matrices.
2. Compute the need matrix.
3. Check whether the system is in a safe state.

PROGRAM

alloc = [[0, 1, 0], [2, 0, 0], [3, 0, 2]] maxm = [[7, 5, 3], [3, 2, 2], [9, 0, 2]] avail = [3, 3, 2]

```
n = len(alloc) m = len(avail)
need = [[maxm[i][j] -
alloc[i][j] for j in range(m)]:
    for i in range(n)]:
        finish = [False] * n
    safe_seq = []
while len(safe_seq) < n: found =
  False for i in range(n):
if not finish[i] and all(need[i][j] <= avail[j] for
j in range(m)):
 for j in range(m):
        avail[j] +=
         alloc[i][j]
  safe_seq.append(i)
    finish[i] = True
    found = True
if not found:
  break

if len(safe_seq) == n:
  print("System is in safe
  state")
print("Safe sequence:", safe_seq)
  else: print("System is not in
  safe state")
```

OUTPUT

Practice 3 - Sample Output

python output

```
System is in safe state
Safe sequence: [1, 2, 0]
```

RESULT

Thus, the given programs were implemented successfully.

The expected outputs were obtained and the required concepts were verified.